

Project 4 – Password Cracking with JTR

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Background

Install John the Ripper from the Fedora Repository and use it to crack a weak password of a newly created user.

Objectives

1. Start Fedora server virtual machine.
2. Connect your VM to the internet.
3. Download JTR from the Fedora repo.
4. Create a new user.
5. Set password to a weak password.
6. Use cat & grep to filter account from shadow file.
7. Run john to crack password.
8. Document steps for inclusion in a project report to be submitted for review and grading in Blackboard.

Prerequisites

1. A Windows or MacOS device with a functioning hypervisor (e.g., VMware Workstation Pro or VMware Fusion Pro).

Connect your VM to the yellow network

1. Start your VM and log in as root.
2. Go to the VM option in the tab and click on settings.
3. Change the network adapter from NAT to custom and select the yellow option.
4. Type `ifconfig` . Notice that the IP address is not automatically assigned to your VM.
5. Type the following commands:

```
sudo nmcli connection modify ens160 ipv4.method auto  
sudo nmcli connection down ens160; sudo nmcli connection up ens160
```
6. Type `ifconfig` again to check if the IP address was now assigned. [Add screenshots.](#)

Install John the Ripper Package and Create New User

7. Now type the following command

```
yum install john
```
8. Create a new user named `jtrtest` and set the password to `abc123`. Show the commands below in a [screenshot](#):

Run john to crack the weak password

JTR contains a program named `unshadow` that combines the contents of the `passwd` and `shadow` files:

9. `cd ~`

10. `unshadow /etc/passwd /etc/shadow > mypasswd`

If we tried to crack the passwords in this file, it would take a long time due to the number of accounts contained in the file (all of them) so we want to isolate the account we just created.

11. Using the `cat` and `grep` commands, filter the user account `jtrtest` and create a new file named `cracklist` that contains only the single account `jtrtest`.

a. `cat mypasswd | grep jtrtest > cracklist`

12. Show your command and display the contents of the file below:

13. Run the following command to crack the password of the `jtrtest` account:

`john -format = crypt cracklist`

Show your work

Once complete, you can use the following command to show the results:

13. `john - show cracklist`

Submission

Use this document (or one of your choosing) to submit your screen shots and any other relevant data. At a minimum, you should show the commands run after logging in as root. Please ensure your submission includes your name, class, date, and reference to this lab.